

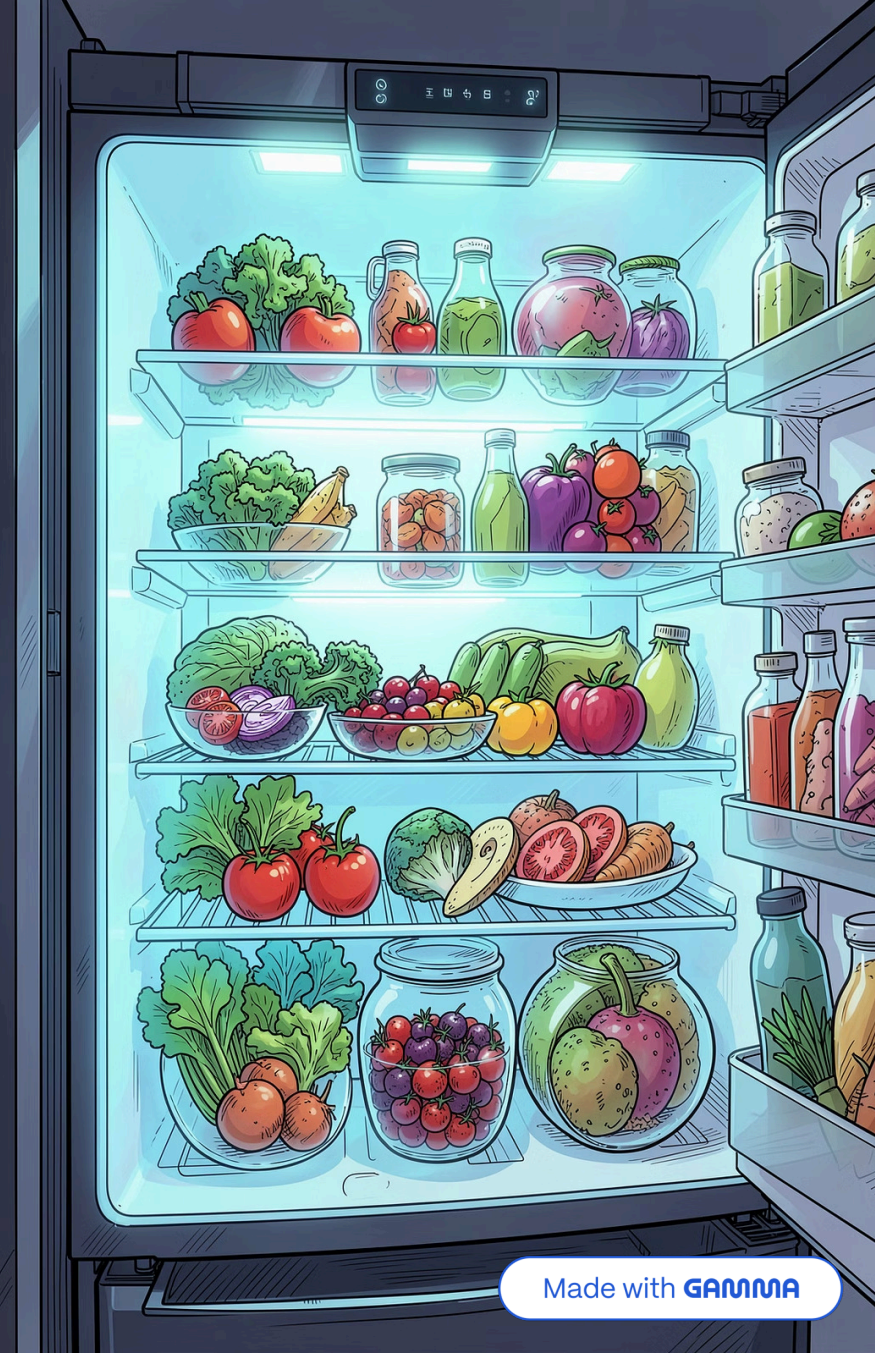
PredictaShelf 🏺

Smart AI-Powered Expiry Tracker

"Never waste food again — AI predicts before it expires"

HACKATHON DEMO

AI / ML



Made with GAMMA

Food Waste Is a Global Crisis

1/3

Food Wasted

Of all food produced globally is
wasted every year

\$1T

Annual Loss

Estimated global economic cost of
food waste

0

Smart Warnings

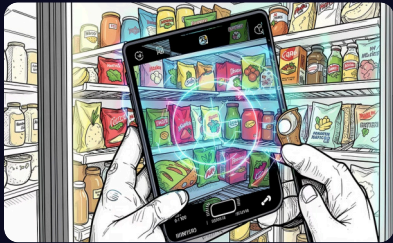
Existing systems that alert users
before expiry

"Out of sight, out of mind — until it's already expired."



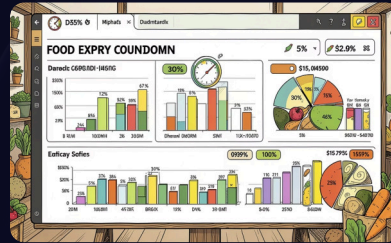
Introducing PredictaShelf

A smart, AI-powered system that scans your fridge, identifies food items, predicts expiry dates, and alerts you — **before it's too late.**



Snap & Detect

Upload a fridge photo — Gemini Vision identifies every item automatically.



AI Predicts Expiry

4 ML models work together to predict exact days remaining per product.

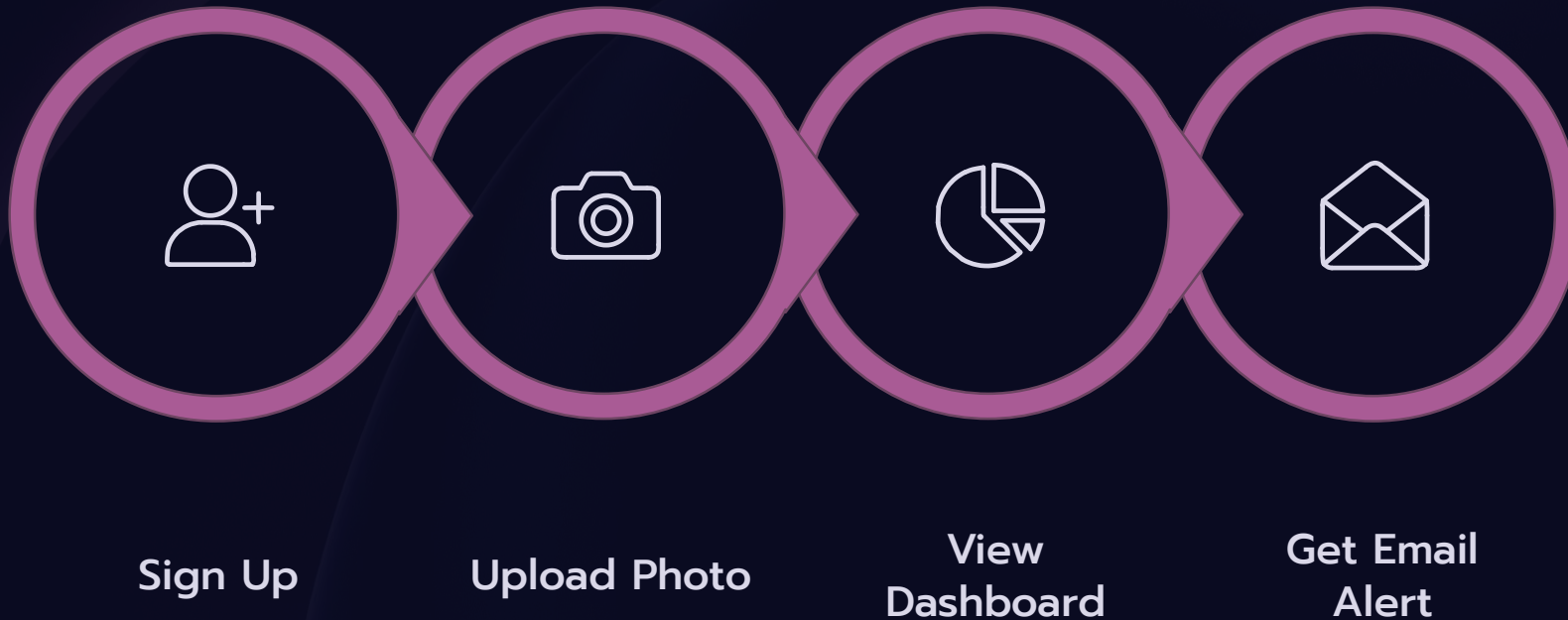


Automated Alerts

Gmail notifications sent daily — zero effort required from the user.

A Simple 4-Step Experience

No manual entry. No guesswork. Just click, submit, and let AI handle the rest.

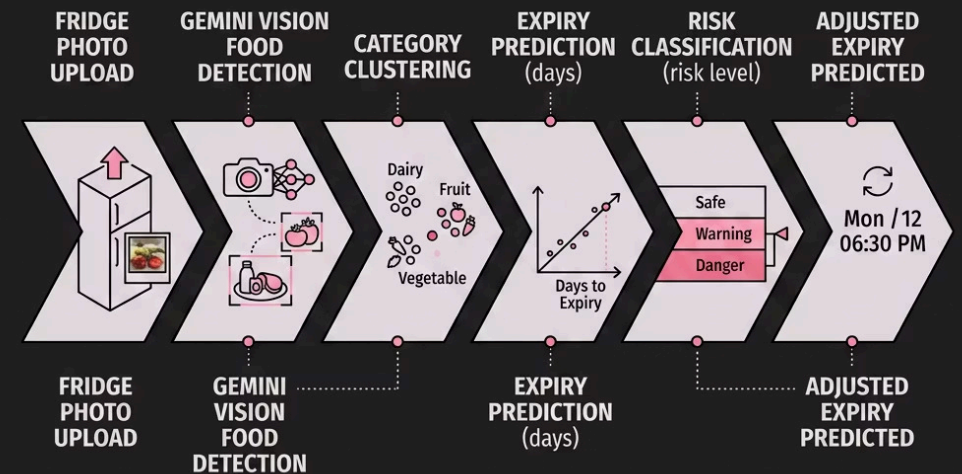


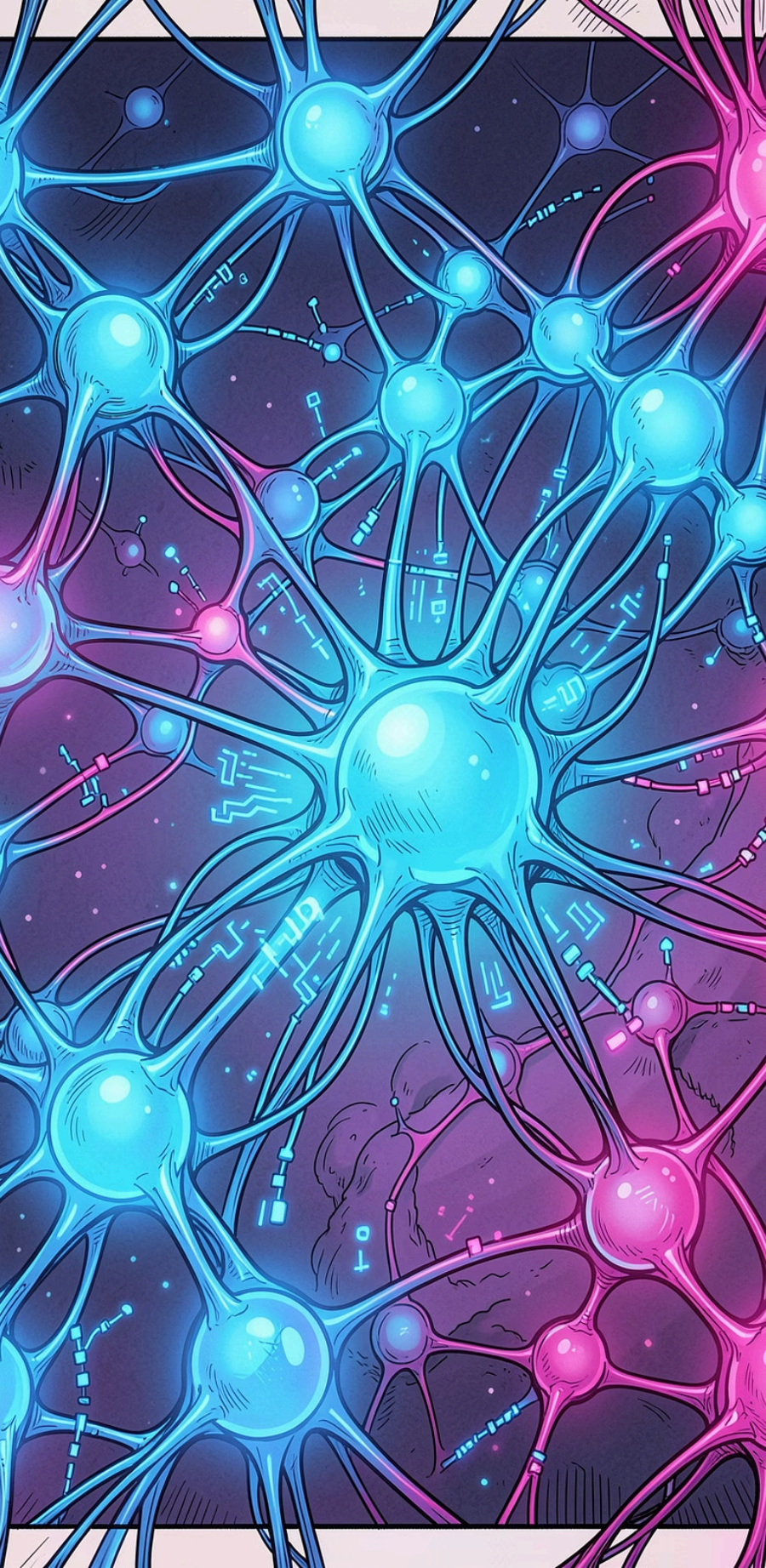
From signup to alert in under 60 seconds — PredictaShelf removes all friction from food tracking.

The AI Pipeline — Core Innovation

PredictaShelf's prediction engine chains **4 AI models** into a single intelligent pipeline. Each layer adds accuracy and a fallback — ensuring a reliable expiry date even when one model is uncertain.

i No single point of failure. If one model can't predict, the next takes over.





4 Models, One Smart System

1

Gemini Vision (Google AI)

Reads fridge images and identifies every food item — no manual labels needed.

2

Linear Regression

Predicts the exact number of days remaining before a product expires.

3

Logistic Regression

Classifies expiry risk into three tiers: Safe, Warning, or Danger.

4

Time Series Analysis

Adjusts predictions dynamically based on freshness signals over time.

Automated Gmail Alert System



PredictaShelf sends **daily automated emails** directly to each user's registered Gmail — with zero action required. Each alert is a polished HTML email with product names, categories, and a direct dashboard link.

 **Already Expired**

Discard immediately

 **Expires Today**

Use right now

 **1–3 Days Left**

Plan your meals

Tech Stack

PredictaShelf is built on a modern, production-ready stack — combining Python's ML ecosystem with Google's vision AI and a robust relational database.

Frontend	HTML, CSS, JavaScript + Jinja2 Templates
Backend	Python — Flask Framework
Database	PostgreSQL (per-user product storage)
AI Vision	Google Gemini Vision API
ML Models	scikit-learn — Linear Reg, Logistic Reg, Clustering, Time Series
Email	Python smtplib + Gmail SMTP Auth
Languages	English  · Hindi  · Gujarati 

What Makes PredictaShelf Different



AI Image Scan

Zero manual data entry — just snap a photo and AI does the rest.



4 ML Models in One Pipeline

Layered prediction engine with built-in fallbacks for maximum accuracy.



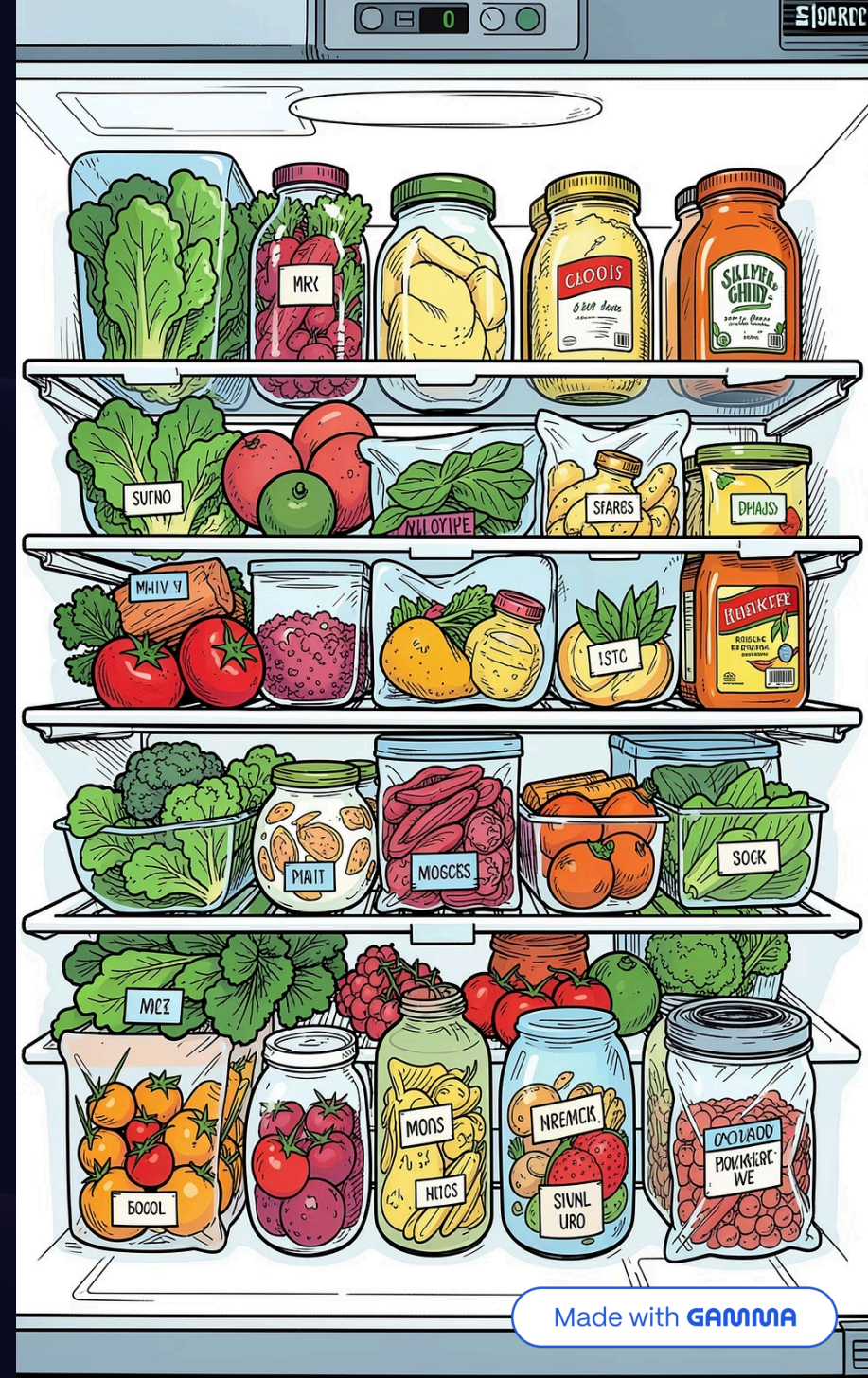
Fully Automated Alerts

Daily Gmail notifications — users never have to check manually.



Multilingual Support

Works in English, Hindi, and Gujarati — accessible to everyone.





The Future of Smart Kitchens



Reduce Waste

AI-powered predictions cut household food waste at the source.



Save Money

Users know exactly what to use first — reducing unnecessary purchases.



Scalable

Built to expand from homes to grocery stores, restaurants, and warehouses.

"We didn't just build an app. We built a system that thinks ahead — so your food doesn't go to waste." 🍲